

THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 6

# The Unreasonable Effectiveness Problem

Note Template

Self-Study Curriculum

**How to use this template**

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

**1. The core question**

*state it in your own words, not Wigner's phrasing*

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**2. The “easy” part of the answer (not actually mysterious)**

a.

b.

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**3. The “hard” part (genuinely mysterious)**

*give one example of math developed purely abstractly that turned out to be physically necessary*

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#### 4. The four structural answers

- a. Structural constraint:
- b. Compression:
- c. Symmetry:
- d. Selection:

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#### 5. Which of these answers I find most compelling, and why

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#### 6. What remains genuinely open

*state it precisely, what exactly is the unsolved part?*

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#### 7. How the compression argument connects to my earlier intuition

*your words*

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## 8. My current best answer to “why does math work?”

*a paragraph: it doesn't have to be final*

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